Weekly Teaching Plan (AIML Program) : Week 05

🟢 **Monday**

🕘 **9:30 – 11:00**

Introduction to Machine Learning (Part 1)

- What is Machine Learning?
- AI vs ML vs Deep Learning
- Real-world applications (industry use-cases)
- Types of ML:
    - Supervised Learning
    - Unsupervised Learning
    - Semi-supervised Learning
    - Reinforcement Learning (basic idea)

🕚 **11:15 – 1:00**

Introduction to Machine Learning (Part 2)

- ML problem types:
    - Classification vs Regression
- ML pipeline overview:
    - Data collection
    - Data preprocessing
    - Model training
    - Evaluation
    - Deployment
- ML project lifecycle (end-to-end)
- Common challenges in ML projects

🕑 **2:00 – 5:00**

🚫 **IIT Faculty Session (No external session scheduled) : Building Interpretable models: Decision Trees & Splitting Mathematics**

🟢 **Tuesday**

🕘 **9:30 – 11:00**

EDA (Exploratory Data Analysis) – Part 1

- What is EDA and why it is important
- Steps involved in EDA
- Data understanding using Descriptive statistics
- Handling missing values (null value imputation)

🕛 **11:15 – 1:00**

EDA – Part 2 (Visualization)

- Univariate analysis:
  - Histograms, boxplots
- Bivariate analysis:
  - Scatter plots, correlation
- Multivariate analysis
  - Heatmaps, Pair-plot, scatterplots
- Identifying outliers
- Understanding distributions (normal/skewed)

🕑 **2:00 – 3:30**

EDA – Part 3 (Hands-on)

- Dataset walkthrough (real dataset)
- Identifying patterns and trends
- Feature relationships
- Business insights from EDA

🕒 **3:45 – 5:00**

EDA – Part 4

- Feature importance intuition
- Detecting multicollinearity
- Creating EDA summary report
- Common EDA mistakes

🟢 **Wednesday**

🕘 **9:30 – 11:00**

Data Preprocessing – Part 1

- Importance of preprocessing
- Data cleaning:
    - Missing values (mean/median/mode, drop)
    - Duplicate handling
- Outlier treatment:
    - IQR method
    - Z-score method

🕚 **11:15 – 1:00**

Data Preprocessing – Part 2

- Encoding techniques with coding implementation:
    - Label Encoding
    - One-Hot Encoding
    - Ordinal Encoding
    - Frequency Encoding
    - Target Encoding (advanced)
- When to use which encoding

🕑 **2:00 – 3:30**

Data Preprocessing – Part 3

- Feature scaling with coding implemenation:
    - Min-Max Scaling
    - Standardization

- When scaling is required
- Log / Power Transformations
    - Log Transformation
    - Square root transformation
    - Box – cox transformation
- Train-test split

🕒 **3:45 – 5:00**

Data Preprocessing – Part 4 (Hands-on)

- End-to-end preprocessing pipeline
- Applying all steps on dataset
- Preparing final dataset for modeling

🟢 **Thursday**

🕘 **9:30 – 11:00**

Linear Regression – Part 1 (Concepts)

- What is Linear Regression
- Mathematical intuition
- Simple Linear Regression
- Explanation of OLS and Gradient descent algorithms in Linear regression
- Assumptions of Linear Regression

🕐 **11:15 – 1:00**

Linear Regression – Part 2 (Implementation)

- Implementation using Python
- Multiple Linear Regression
- Interpretation of coefficients

🕑 **2:00 – 3:30**

Linear Regression – Part 3

- Polynomial Regression
- Overfitting vs Underfitting
- Model complexity

🕒 **3:45 – 5:00**

Linear Regression – Part 4

- Evaluation metrics:
  - MAE, MSE, RMSE, MAPE
  - R^2 score, Adjusted R^2 score
- **Case Study (end-to-end)**:
  - Build model → evaluate → interpret

🟢 **Friday**

🕘 **9:30 – 11:00**

Bias-Variance Tradeoff

- What is Bias
- What is Variance
- Underfitting vs Overfitting
- Tradeoff intuition

- Model generalization

🕐 **11:15 – 1:00**

Logistic Regression – Part 1 (Concepts)

- Classification vs Regression recap
- What is Logistic Regression
- Sigmoid function
- Decision boundary

🕑 **2:00 – 3:30**

Logistic Regression – Part 2 (Implementation)

- Code implementation
- Binary classification example
- Model training

🕒 **3:45 – 5:00**

Logistic Regression – Part 3

- Evaluation metrics:
    - Accuracy
    - Precision, Recall, F1-score
    - Confusion Matrix
- **Case Study (end-to-end)**